

Exploring the Use of Large Language Models for Improving the Awareness of Mindfulness

Harsh Kumar
University of Toronto
Toronto, Ontario, Canada
harsh@cs.toronto.edu

Yiyi Wang
University of Toronto Mississauga
Mississauga, Ontario, Canada
yiyi.wang@mail.utoronto.ca

Jiakai Shi
University of Toronto
Toronto, Canada
jiakai.shi@mail.utoronto.ca

Ilya Musabirov
University of Toronto
Toronto, Canada
imusabirov@cs.toronto.edu

Norm A.S. Farb
University of Toronto Mississauga
Mississauga, Ontario, Canada
norman.farb@utoronto.ca

Joseph Jay Williams
University of Toronto
Toronto, Canada
williams@cs.toronto.edu

2 (Informational Chatbot: present vs. absent) x 2 (Tutorial Video: present vs. absent) x 2 (Reflection Chatbot: present vs. absent)

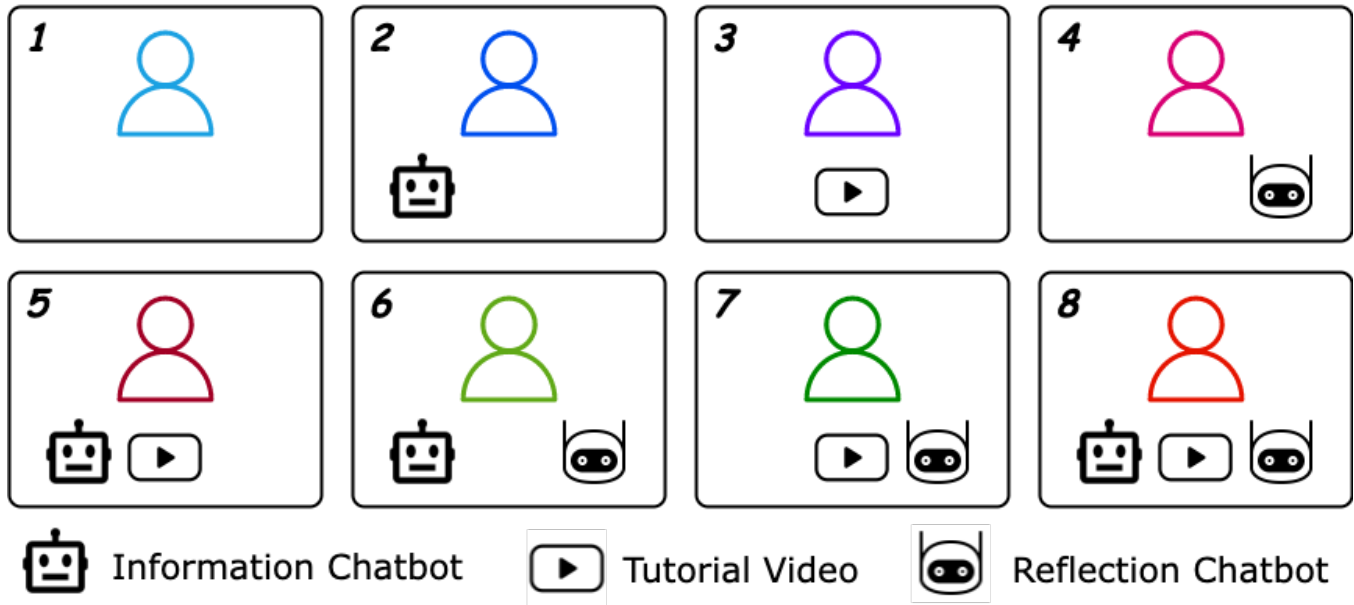


Figure 1: Overall design of the experiment - 2x2x2 factorial design for Information Chatbot, Tutorial Video and Reflection Chatbot, with participants assigned to one of the conditions for a between-subjects design.

ABSTRACT

Teachable self-help techniques, such as mindfulness, can reduce anxiety and improve mental well-being outcomes. However, people lack proper awareness of such techniques. In this work, we explore the design space of using online dissemination channels to help

people learn about mindfulness. We investigate the potential benefits of using Large Language Models (LLMs) to improve awareness and willingness to practice these techniques, building on a video-based intervention to introduce mindfulness. We designed a pilot between subjects randomized factorial experiment of 2 (Informational Chatbot: present vs. absent) x 2 (Tutorial Video: present vs. absent) x 2 (Reflection Chatbot: present vs. absent). Our preliminary findings suggest that interaction with either of the chatbots improved the participants' intent to practice Mindfulness again, and the tutorial video improved the participants' reported overall experience of the exercise. This highlights the potential promise and outlines the directions for exploring the use of LLM-based chatbots for awareness-related interventions.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '23, April 23–28, 2023, Hamburg, Germany
© 2023 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3585614>

CCS CONCEPTS

• **Human-centered computing** → **Natural language interfaces**; *Interaction design*; *Laboratory experiments*; *Field studies*.

KEYWORDS

mental well-being, GPT-3, factorial experiment, large language models

ACM Reference Format:

Harsh Kumar, Yiyi Wang, Jiakai Shi, Ilya Musabirov, Norm A.S. Farb, and Joseph Jay Williams. 2023. Exploring the Use of Large Language Models for Improving the Awareness of Mindfulness. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585614>

1 INTRODUCTION

Mindfulness is a quality, while meditation is a practice [20]. Kabat-Zinn [20, 21] defines mindfulness as non-judgmental awareness of the present moment. To cultivate mindful awareness, attention must be combined with a non-judgmental orientation towards and openness to the flow of one's experiences [27]. Mindfulness is associated with many benefits to mental health, as well as other positive attributes [38]. Teachable mindfulness techniques can reduce anxiety [19, 22], promote awareness of thoughts, emotions, and sensations [4, 15], aid in the development of adaptive regulatory strategies [2, 14], and cultivate goals and action commitment [10, 39].

However, people lack proper awareness of improving their mental well-being by practicing techniques such as mindfulness. Increasing awareness of such techniques beyond traditional therapeutic settings is an important task. Recently, developments in large language model (LLM)-based conversational agents reopen the design space to explore whether designers can use conversational agents as intervention components to increase awareness and readiness to engage in mental well-being techniques [5, 9].

In this work, we explore the design space of using online dissemination channels to help people learn about mindfulness. We investigate the potential benefits of using chatbots based on large language models (LLMs) to improve awareness and readiness to engage, building on an established video-based intervention [38].

Participants recruited through Amazon Mechanical Turk were evenly divided under different conditions of a factorial experiment of 2 (*Informational Chatbot*: present vs. absent) $\times$ 2 (*Tutorial Video*: present vs. absent) $\times$ 2 (*Reflection Chatbot*: present vs. absent) for an ablation-style study to investigate the complex combination of all these different elements, varying whether or not one of these components is present. Our findings indicate that both chatbots, the one programmed to provide information related to Mindfulness (InformationalChatbot) and the other initialized to help participants reflect, improved the participants' ratings of their intent to practice Mindfulness again after undergoing the exercises. The tutorial video showed an improvement in the self-reported "overall experience" of the participants.

We found that the introduction of any kind of interaction (tutorial video or chatbot) improved the participants' intent to practice mindfulness again, and increased their overall experience of the whole exercise, compared to the baseline condition where only a

text about Mindfulness was shown. We also found that the number of participants who rated the lowest score for their overall experience is 0 when introducing any type of chatbot (see Section 4. This highlights the promise of exploring the use of LLM-based chatbots for awareness-related interventions.

2 RELATED WORK

In this section, we look at some existing literature related to mindfulness and web-based interventions, followed by a look at some applications of LLMs.

2.1 Mindfulness and Web-based Interventions

The Mindfulness to Meaning Theory (MMT) proposes a practical approach to human flourishing by combining insights from mindfulness and cognitive reappraisal [17]. Rather than simply focusing on outcomes such as flourishing or stress reduction, MMT emphasizes the development of specific skills, such as decentering and positive reappraisal, which improve well-being [18]. MMT recognizes the value of mindfulness and stress mindset, but suggests that these strategies work best in combination, with decentering providing flexibility in appraisal and positive re-appraisal providing meaning and purpose. There is a limited amount of research on the combined effects of decentering and positive reappraisal, but each has independent benefits.

The use of online mindfulness interventions has grown rapidly in the past decade, with a recent meta-analysis identifying 97 online mindfulness-based interventions [35]; the analysis suggested that online MBIs show moderate effects on depression, stress, and dispositional mindfulness with small effects on anxiety and well-being. Mak et al. [26] implemented an 8-week internet-based mindfulness intervention in university students and found significant improvements in mindfulness and mental well-being of students. Similarly, students who reported a more frequent use of app-based mindfulness meditation (that is, Headspace) demonstrated improved psychological symptoms and college adjustment [16]. Successful implementation of online and app-based mindfulness interventions strengthened the possibility of using online mindfulness interventions to support students during the pandemic.

Digital tools (e.g., computers and mobile phones) provide a method of supporting people's mental health outside of clinical settings [23]. In a recent online mindfulness study, researchers successfully supported students' mental health and coping strategies with brief online mindfulness interventions [38].

2.2 Application of Large Language Models

LLMs have been applied in a wide range of contexts within the HCI community to design assistive technologies that generate human-like logical and intellectual responses to prompts [5, 9]. These models have been extensively applied to support writing tasks [33] such as story writing [42], user-adapted semantic description generation [34] and help curate emails [6]. In other domains, such as helping users with their mental health, researchers have shown the potential of using LLMs for improving the mood of users [24] and help computer science undergraduate students manage their mental well-being through interactions with chatbots [25]. Models

like GPT-3 [5] can be "prompted" with few examples to produce coherent results for various use cases [13].

3 EXPERIMENT DESIGN AND METHODS

In this study, we conducted a pilot experiment to explore ways to answer the following questions: 1) How do LLM-based chatbot interventions compare to existing video-based methods for improving mental health awareness; 2) Are LLM-based chatbots increasing the effectiveness of existing video-based awareness methods; and 3) What types of LLM-based chatbot are most effective in different contexts.

To answer these, we designed a randomized factorial experiment of 2 (*InformChatbot*: present vs. absent) x 2 (*InformVideo*: present vs. absent) x 2 (*ReflectionChatbot*: present vs. absent) between subjects. We used an ablation-style study, where we investigated the complex combination of all these different elements by varying whether or not one of these components is present. The design of the experiment is shown in Figure 1.

We recruit participants from Amazon Mechanical Turk (AMT), which has been used frequently in social and behavioral science studies [3, 7]. We use GPT-3 carefully programmed using the zero-shot learning approach to design both chatbots in the study [5, 24, 32], described in detail in Sections 3.1 and 3.3.

3.1 Factor 1 - Informational Chatbot

The purpose of this chatbot is to provide information related to mindfulness to the participant through dialogues. Computer programs designed to replicate the teaching of human teachers have been studied for decades [8, 36, 37]. LLMs have shown potential in supporting in-context few-shot learning through *prompting* - one can prompt the model with few input and output exemplars of the desired task, and the model predicts the next sequence of responses [5, 9, 24, 40, 43]. Based on the previous literature related to prompting [13, 24], learning [11], and repeated tests, we designed the following input prompt for the GPT-3 model:

The following is a conversation with a Mindfulness instructor. The instructor teaches and provides information about different mindfulness activities to the Human. The instructor explains different activities clearly and provides examples wherever possible. The instructor has a sense of humor, is fair, and is empathetic.

We use OpenAI guidelines to ensure that the prompt includes an identity, intent, and behavior of the model while generating responses [1]. Along with the input prompt, the model was initialized with the first message from the bot to the user that said *"Hello. I am an AI agent designed to act as your Mindfulness instructor. I can answer any questions you might have related to Mindfulness. How can I help you?"*. Participants were shown some sample prompts they could use to start the conversation, such as *"Can you help me learn about Mindfulness?"*. Section 4.1 shows a conversation one of the study participants had with a chatbot initialized with this prompt.

3.2 Factor 2 - Tutorial Video

The training video was adapted from an existing study [38], which was based on mindfulness-based interventions, introducing participants to mindfulness meditation, and featuring exercises to focus attention on the internal sensation of the body or breath with a curious, non-judgmental attitude. Because the main objective of the study was to test whether we can introduce people to the world of mindfulness, participants were shown only one training video, mindful breathing, which is one of the most fundamental mindfulness sessions.

3.3 Factor 3 - Reflection Chatbot

The purpose of this chatbot is to help participants reflect on their knowledge of mindfulness, and plan when they would perform Mindfulness exercises daily. Existing research has shown an improvement in learning outcomes for students when asked to reflect on their metacognitive strategies and beliefs [31, 41]. To enable this reflection, we follow a similar procedure mentioned in Section 3.1 to program a GPT-3 based chatbot using the following prompt:

The following is a conversation with a Mindfulness instructor. The instructor asks open-ended reflection questions to the Human to solidify the Human's understanding of Mindfulness and helps them plan when they can practice Mindfulness in their daily lives. The instructor has a sense of humour, is fair, and empathetic.

Furthermore, we add the first message from the bot to the initial prompt saying *"Hello. I am an AI agent designed to act as your Mindfulness instructor. I am here to help you reflect on your learnings. How can I help you?"*, which was also displayed to the participants. We also provided some sample messages to participants that they could use to initialize the dialogue with the chatbot (for example, *"Can you help me plan when I can practice Mindfulness?"*). Section 4.1 shows an example conversation one of the participants had with this chatbot.

3.4 Participants

We obtained 209 responses to our Qualtrics survey through participants recruited from Amazon Mechanical Turk [12]. After filtering out random respondents with our attention checks, we had 189 valid responses. The demographic information of the participants is shown in Table 1. The participants came from a variety of backgrounds and most had some previous experience with mindfulness meditation (93.12%). Each participant was randomly assigned to one of the eight (2x2x2) experimental conditions (see Section 3), where they spent a maximum of 20 minutes doing the assigned activities and 10 minutes answering the follow-up questions.

3.5 Procedures

Participants from MTurk were directed to a Qualtrics survey [30]. They were informed about the purpose of the study, to explore and compare various ways of introducing mindfulness and eventually helping people lead happier lives. They were also provided with contact information for crisis management centers in case they felt emotional distress while completing the survey (See Section

Total Participants = 189	
Age	18-25 (9.52%), 25-34 (48.68%), 35-44 (22.22%), 45-54 (14.29%), 55-64 (3.70%), 65+ (1.59%)
Gender	Male (53.44%), Female (46.56%)
Employment	Full-time (95.24%), Part-time (1.06%), Student (1.06%), Retired (2.64%)
Marital Status	Married/cohabitant (79.89%), Partner but not living together (0.53%), Single/divorced (19.58%)
Education	High School graduate (2.65%), Some college credit with no degree (4.76%), Associate degree (1.59%), Bachelor's degree (66.67%), Master's degree (20.63%), Professional degree (2.64%), Doctorate degree (1.06%)
Annual Income	Less than \$10K (1.06%), 10-49K (33.33%), 50-99K (54.50%), 100K+ (11.11%)
Ethnicity	White/Caucasian (84.66%), Black/African American (2.12%), American Indian/Native American or Alaska Native (1.06%), Asian (11.11%), Prefer not to say (0.53%)
Mindfulness Experience	Yes (93.12%), No (6.88%)

Table 1: Demographic information of the participants.

3.7). We also implemented multiple attention checks to filter out low-quality responders.

3.6 Dependent Variables

Dependent variables were measured after the participant completed all assigned exercises (see Section 3).

Overall experience Participants were randomly assigned to one of eight (2x2x2) experimental conditions after which they were asked to rate, on a scale of 1 (Terrible) to 7 (Excellent), their overall experience of undergoing this exercise.

Intent to practice mindfulness Since our focus was to increase awareness of Mindfulness and increase the chances that participants practice mindfulness on their own after undergoing this exercise, they were asked to rate on a scale of 1 (not likely at all) to 7 (extremely likely) "How likely are you to practice Mindfulness after undergoing these learning exercises?".

3.7 Ethical Considerations

Research related to mental health raises several ethical issues, the most notable being the fact that experimental interventions can induce negative emotions, whether intentionally during self-reflection or unintentionally due to misinterpretation of information. To mitigate this risk, we warn crowdworkers that they may encounter disturbing content and actively inform them of their right to stop completing the survey. They were also provided key

mental health resources at the beginning of the survey (e.g., The National Suicide Prevention Lifeline information).

4 PRELIMINARY RESULTS

In this section, we report descriptive results of the comparison, informing further larger-sample confirmatory study.

4.1 Descriptive Quantitative Results

Figure 2 A and B show the distribution of % of participants by ratings for each factor and their levels.

Informational Chatbot We observe a decrease in the number of participants who indicated they were "Not likely at all" to practice Mindfulness exercises after undergoing the exercises for participants who were assigned to the InformationChatbot condition (Figure 2A). Interaction with InstructionChatbot does not seem to largely affect the positive part of the spectrum of ratings. For the overall experience, differences appear to be inside each part of the spectrum without apparent changes in the proportions of positive and negative ratings.

Tutorial Video Participants assigned to the TutorialVideo condition showed a slight increase in positive ratings (score of 5, 6 or 7 out of 7) when asked about their overall experience, compared to those who were not assigned to the video arm. The apparent effect appears to be less pronounced for the participants' intention to practice mindfulness again.

Reflection Chatbot For participants assigned to the Reflection-Chatbot condition, we observe a notable decrease in participants who rated the lowest possible value ("Not likely at all") and an increase in the number of participants who gave a positive score for intention to practice mindfulness again (Figure 2A). This is in line with existing research on the role of reflection in promoting behavior change [28, 29]. Interaction with ReflectionChatbot did not have a notable effect on the overall experience of the participants.

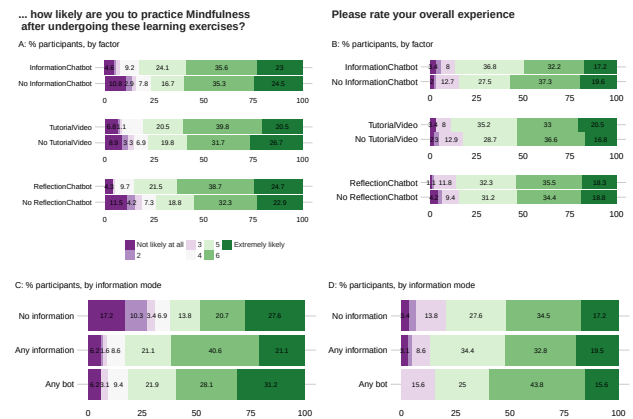


Figure 2: Overall summary of the distribution of ratings for each dependent variables.

Another formative question is how the introduction of any bot or information compares to the baseline condition of no information. We combine the observations in the following manner. "No information" indicates the condition without InstructionChatbot,

TutorialVideo, and ReflectionChatbot (see Figures 2 C and D). In this condition, the participant only gets a block of text that describes Mindfulness and its benefits, representative of how a lot of information is shared on social media through text. This gives us the opportunity to explore the use of videos and chatbots to increase the effectiveness of these messages. "Any information" combines all observations which had at least a video or one of the chatbots. "Any bot" includes observations that had at least one bot but no video.

Intent to practice Mindfulness again Percentage of participants who said they were "Not likely at all" to practice mindfulness again is notably decreased with the introduction of any form of information (video or chatbot). The number of participants who provided a positive score also increased with the introduction of any form of information/interaction. This suggests that there is a potential promise of using chatbots or videos for awareness-related interventions to increase the chances of positive behavior change in participants.

Overall experience Participants in "Any information" conditions reported an increase in positive ratings for their overall experience, compared to participants in "No information" condition. The introduction of any type of bot (InstructionChatbot or ReflectionChatbot) reduced the number of participants who gave the lowest rating to 0.

4.2 Qualitative Experience Description

InformationalChatbot vs ReflectionChatbot We see qualitative differences in the dialogues generated by both bots. For InstructionChatbot, the bot generated 299 words per session on average, compared to the participant typing 35 words on average. This is in contrast to the ReflectionBot, where the bot generated comparatively less words (255) but elicited more responses from the user (44 words) on average.

The following is an example conversation one of the participants had with ReflectionChatbot. The bot was programmed using the prompts in 3.3.

👤 What situations in life could I use mindfulness?

🗣️ You can use mindfulness in any situation where you want to be more present and aware. For example, you might use mindfulness when you're eating, walking, or interacting with others. When would you like to try using mindfulness?

👤 I think it could be good in stressful situations. I think it could help ease the stress and calm my nerves.

🗣️ That sounds like a great idea! Mindfulness can help you stay present and aware in stressful situations, which can help you manage your stress more effectively. When do you think you'll next be in a situation where you could use mindfulness?

👤 I have to take my husband to a physical therapy session, which is hard on him. Do you think it would help in that situation?

🗣️ Mindfulness can definitely help in that situation! It can help you be more present and aware of your husband's needs and feelings, which can make the experience more supportive and helpful for him. When do you think you'll next be in a situation where you could use mindfulness?

👤 Perhaps when I have a meeting with my boss.

🗣️ That's a great idea! Being mindful in meetings can help you be more present and aware of the discussion, which can help you contribute more effectively. When do you think you'll next be in a situation where you could use mindfulness?

👤 Well, what about when I am around loud children? Is that a situation that it wouldn't necessarily be helpful?

🗣️ There is no right or wrong answer when it comes to using mindfulness – it is entirely up to you and what you feel would be helpful in any given situation. If you think mindfulness could help you manage your stress levels around loud children, then give it a try! There is no harm in doing so.

The following is a sample conversation from a participant who interacted with the InformationalChatbot in the study 3.1.

👤 Let's do some mindfulness!

🗣️ Sounds great! There are many different mindfulness activities you can do. For example, you can focus on your breath, or you can focus on a particular object. You can also do a body scan, where you focus on each part of your body from head to toe.

👤 Where do we start?

🗣️ We can start with a breathing exercise. Breathing is a great way to calm the mind and focus on the present moment.

👤 Let's go!

🗣️ Okay, let's start by focusing on your breath. Simply close your eyes and focus on the sensation of your breath entering and leaving your body. Don't try to control your breath, simply observe it.

👤 Done.

🗣️ How was that?

👤 Good.

🗣️ I'm glad you found it helpful! Mindfulness can be a great way to reduce stress and focus on the present moment.

5 CONCLUSION AND FUTURE WORK

In this work, we explored the design space of using online dissemination channels to help people learn about Mindfulness. We investigated the potential benefits of using Large Language Models (LLMs) based chatbots to improve the awareness and readiness to engage, building on an established video-based intervention. In summary, we explore three key insights. First, we investigate the value of a tutorial video showing users how to do mindful breathing exercises. Second, we investigate the potential role of information related to mindfulness being explained via chatbot. Third, using another chatbot to encourage people to reflect on what they know about mindfulness and plan when to practice it next.

Through this formative study, we collected a dataset of qualitative responses to several questions to derive design insights for future deployments. We also collected a repository of conversation logs between participants and two differently programmed GPT-3 chatbots. All of this will help inform the next set of larger-scale deployments. Our work relied on using a survey for participants recruited from Mechanical Turk that raises questions about the representativeness of the sample and the quality of the responses. For example, 93.12% of the participants reported some previous experience with mindfulness, which requires diversification of the

sample and taking into account previous experience. We also plan to conduct semi-structured interviews to have richer exploratory studies on what the participants with and without the previous mindfulness experience see as the main value of different chatbot modes. Among the explorations is also the question of whether and how the interaction with different chatbot modes differs in the eyes of the participants.

ACKNOWLEDGMENTS

We would like to thank Huayin Luo, Ege Mert Akin, Xinyuan Wang, Harry Zhang, Raymond Zhang, and Yi Wang for helping with the preprocessing of the data and proofreading the manuscript. We also thank Mohi Reza and the reviewers for their detailed feedback which helped to improve this manuscript. We acknowledge the support of the Natural Sciences and Engineering Research Council of Canada (NSERC) (#RGPIN-2019-06968) and the Office of Naval Research (ONR) (#N00014-21-1-2576) grants awarded to Dr. Williams, and the SSHRC Insight Grants (43520220146) awarded to Dr. Farb. In addition, we acknowledge a gift from the Microsoft AI for Accessibility program that, in part, supported this work (<http://aka.ms/ai4a>).

REFERENCES

- [1] Open AI. 2022. Open AI API Documentation. <https://beta.openai.com/docs/guides/completion/prompt-design>
- [2] Alon Alkoby, Ruthie Pliskin, Eran Halperin, and Nava Levit-Binnun. 2019. An eight-week mindfulness-based stress reduction (MBSR) workshop increases regulatory choice flexibility. *Emotion* 19, 4 (2019), 593.
- [3] Frank R Bentley, Nediya Daskalova, and Brooke White. 2017. Comparing the reliability of Amazon Mechanical Turk and Survey Monkey to traditional market research surveys. In *Proceedings of the 2017 CHI conference extended abstracts on human factors in computing systems*. 1092–1099.
- [4] Elisabeth S Blanke, Mirjam J Schmidt, Michaela Riediger, and Annette Brose. 2020. Thinking mindfully: How mindfulness relates to rumination and reflection in daily life. *Emotion* 20, 8 (2020), 1369.
- [5] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems* 33 (2020), 1877–1901.
- [6] ERIN BUEHLER, PATRICK CLARY, ANDY COENEN, AARON DONSACH, TIFFANIE N HORNE, MICHAL LAHAV, ROBERT MACDONALD, RAIN BREAW MICHAELS, AJIT NARAYANAN, JOEL RILEY PUSHKARNA, et al. 2022. LaMPPost: Design and Evaluation of an AI-assisted Email Writing Prototype for Adults with Dyslexia. (2022).
- [7] Michael Buhrmester, Tracy Kwang, and Samuel D Gosling. 2016. Amazon's Mechanical Turk: A new source of inexpensive, yet high-quality data? (2016).
- [8] Jaime R Carbonell. 1970. AI in CAI: An artificial-intelligence approach to computer-assisted instruction. *IEEE transactions on man-machine systems* 11, 4 (1970), 190–202.
- [9] Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. 2022. Palm: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311* (2022).
- [10] Kimberly A Coffey and Marilyn Hartman. 2008. Mechanisms of action in the inverse relationship between mindfulness and psychological distress. *Complementary health practice review* 13, 2 (2008), 79–91.
- [11] Bridget Cooper, Paul Brna, and Alex Martins. 2000. Effective affective in intelligent systems—building on evidence of empathy in teaching and learning. In *International Workshop on Affective Interactions*. Springer, 21–34.
- [12] John A Cunningham, Alexandra Godinho, and Vladislav Kushnir. 2017. Using Mechanical Turk to recruit participants for internet intervention research: experience from recruitment for four trials targeting hazardous alcohol consumption. *BMC medical research methodology* 17, 1 (2017), 1–7.
- [13] Hai Dang, Lukas Mecke, Florian Lehmann, Sven Goller, and Daniel Buschek. 2022. How to Prompt? Opportunities and Challenges of Zero- and Few-Shot Learning for Human-AI Interaction in Creative Applications of Generative Models. <https://doi.org/10.48550/ARXIV.2209.01390>
- [14] Norman AS Farb, Zindel V Segal, and Adam K Anderson. 2013. Mindfulness meditation training alters cortical representations of interoceptive attention. *Social cognitive and affective neuroscience* 8, 1 (2013), 15–26.
- [15] Norman AS Farb, Zindel V Segal, Helen Mayberg, Jim Bean, Deborah McKeon, Zainab Fatima, and Adam K Anderson. 2007. Attending to the present: mindfulness meditation reveals distinct neural modes of self-reference. *Social cognitive and affective neuroscience* 2, 4 (2007), 313–322.
- [16] Jayde AM Flett, Tamlin S Conner, Benjamin C Riordan, Tess Patterson, and Harlene Hayne. 2020. App-based mindfulness meditation for psychological distress and adjustment to college in incoming university students: a pragmatic, randomised, waitlist-controlled trial. *Psychology & health* 35, 9 (2020), 1049–1074.
- [17] Eric L Garland, Norman A Farb, Philippe R Goldin, and Barbara L Fredrickson. 2015. The mindfulness-to-meaning theory: Extensions, applications, and challenges at the attention–appraisal–emotion interface. *Psychological Inquiry* 26, 4 (2015), 377–387.
- [18] Judith Glück and Susan Bluck. 2013. The MORE life experience model: A theory of the development of personal wisdom. In *The scientific study of personal wisdom*. Springer, 75–97.
- [19] Ditte Hoffmann, Charlotte Ulrikka Rask, Erik Hedman-Lagerlöf, Jens Søndergaard Jensen, and Lisbeth Frostholt. 2021. Efficacy of internet-delivered acceptance and commitment therapy for severe health anxiety: results from a randomized, controlled trial. *Psychological medicine* 51, 15 (2021), 2685–2695.
- [20] Jon Kabat-Zinn. 1994. Catalyzing movement towards a more contemplative/sacred-appreciating/non-dualistic society. In *Meeting of the working group*.
- [21] Jon Kabat-Zinn. 2003. Mindfulness-based interventions in context: past, present, and future. (2003).
- [22] Bassam Khoury, Tania Lecomte, Guillaume Fortin, Marjolaine Masse, Phillip Therien, Vanessa Bouchard, Marie-Andrée Chapeau, Karine Paquin, and Stefan G Hofmann. 2013. Mindfulness-based therapy: a comprehensive meta-analysis. *Clinical psychology review* 33, 6 (2013), 763–771.
- [23] Rachel Kornfield, Jonah Meyerhoff, Hannah Studd, Ananya Bhattacharjee, Joseph Jay Williams, Madhu Reddy, and David C Mohr. 2022. Meeting Users Where They Are: User-centered Design of an Automated Text Messaging Tool to Support the Mental Health of Young Adults. In *CHI Conference on Human Factors in Computing Systems*. 1–16.
- [24] Harsh Kumar, Ilya Musabirov, Jiakai Shi, Adele Lauzon, Kwan Kiu Choy, Ofek Gross, Dana Kulzhabayeva, and Joseph Jay Williams. 2022. Exploring The Design of Prompts For Applying GPT-3 based Chatbots: A Mental Wellbeing Case Study on Mechanical Turk. <https://doi.org/10.48550/ARXIV.2209.11344>
- [25] Harsh Kumar, Kunzhi Yu, Andrew Chung, Jiakai Shi, and Joseph Jay Williams. 2023. Exploring The Potential of Chatbots to Provide Mental Well-Being Support for Computer Science Students. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 2* (Toronto ON, Canada) (SIGCSE 2023). Association for Computing Machinery, New York, NY, USA, 1339. <https://doi.org/10.1145/3545947.3576285>
- [26] Winnie WS Mak, Amy TY Chan, Eliza YL Cheung, Cherry LY Lin, and Karin CS Ngai. 2015. Enhancing Web-based mindfulness training for mental health promotion with the health action process approach: randomized controlled trial. *Journal of Medical Internet Research* 17, 1 (2015), e3746.
- [27] Adam Moore and Peter Malinowski. 2009. Meditation, mindfulness and cognitive flexibility. *Consciousness and cognition* 18, 1 (2009), 176–186.
- [28] Bernd Ploderer, Wolfgang Reitberger, Harri Oinas-Kukkonen, and Julia van Gemert-Pijnen. 2014. Social interaction and reflection for behaviour change. , 1667–1676 pages.
- [29] Nora Ptauskaitė, Anna L Cox, and Nadia Berthouze. 2018. Knowing what you're doing or knowing what to do: how stress management apps support reflection and behaviour change. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–6.
- [30] Qualtrics. 2002. *Qualtrics*. <https://www.qualtrics.com/>
- [31] Chris Quintana, Meilan Zhang, and Joseph Krajcik. 2018. A framework for supporting metacognitive aspects of online inquiry through software-based scaffolding. In *Educational Psychologist*. Routledge, 235–244.
- [32] Laria Reynolds and Kyle McDonell. 2021. Prompt programming for large language models: Beyond the few-shot paradigm. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–7.
- [33] Bradley Robinson. 2022. Speculative Propositions for Digital Writing Under the New Autonomous Model of Literacy. *Postdigital Science and Education* (2022), 1–19.
- [34] Javier Sevilla Salcedo, Laura Martín Galván, Jose C Castillo, Álvaro Castro-González, and Miguel A Salichs. 2023. User-Adapted Semantic Description Generation Using Natural Language Models. In *International Symposium on Ambient Intelligence*. Springer, 134–144.
- [35] Marion Sommers-Spijkerman, Judith Austin, Ernst Bohlmeijer, Wendy Pots, et al. 2021. New evidence in the booming field of online mindfulness: an updated meta-analysis of randomized controlled trials. *JMIR mental health* 8, 7 (2021), e28168.
- [36] Robert A Sottolare, Arthur Graesser, Xiangen Hu, and Heather Holden. 2013. *Design recommendations for intelligent tutoring systems: Volume 1-learner modeling*. Vol. 1. US Army Research Laboratory.

- [37] Kurt VanLehn. 2006. The behavior of tutoring systems. *International journal of artificial intelligence in education* 16, 3 (2006), 227–265.
- [38] Yiyi Wang and Norman AS Farb. 2022. Web-based training for post-secondary student well-being during the pandemic: a randomized trial. *Anxiety, Stress, & Coping* (2022), 1–17.
- [39] Yiyi Wang, Jonathan Tarbox, Amanda N Chastain, and Michael J Cameron. 2020. The effects of bilingual acceptance and commitment training (ACT) on exercise in bilingual international university students. *Journal of Contextual Behavioral Science* 18 (2020), 1–8.
- [40] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain of thought prompting elicits reasoning in large language models. *arXiv preprint arXiv:2201.11903* (2022).
- [41] Longkai Wu and Chee-Kit Looi. 2012. Agent prompts: Scaffolding for productive reflection in an intelligent learning environment. *Journal of Educational Technology & Society* 15, 1 (2012), 339–353.
- [42] Ann Yuan, Andy Coenen, Emily Reif, and Daphne Ippolito. 2022. Wordcraft: Story Writing With Large Language Models. In *27th International Conference on Intelligent User Interfaces*. 841–852.
- [43] Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Olivier Bousquet, Quoc Le, and Ed Chi. 2022. Least-to-Most Prompting Enables Complex Reasoning in Large Language Models. *arXiv preprint arXiv:2205.10625* (2022).